



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

**SENIOR CERTIFICATE EXAMINATIONS/
NATIONAL SENIOR CERTIFICATE EXAMINATIONS
SENIORSERTIFIKAAT-EKSAMEN/
NASIONALE SENIORSERTIFIKAAT-EKSAMEN**

TECHNICAL MATHEMATICS P1/TEGNIJSE WISKUNDE V1

MAY/JUNE/MEI/JUNIE 2024

MARKING GUIDELINES/NASIENRIGLYNE

MARKS/PUNTE: 150

Marking Codes/Nasienkode	
A	Accuracy/Akkuraatheid
CA	Consistent Accuracy/Volgehoue Akkuraatheid
M	Method/Metode
R	Rounding/Afronding
NPR	No Penalty for Rounding/Geen Penaliserings vir Afronding nie
NPU	No Penalty for Units omitted/Geen Penaliserings vir Eenhede Weggelaat nie
S	Simplification/Vereenvoudiging
SF	Substitution in Correct Formula/Vervanging in Korrekte Formule

**These marking guidelines consist of 19 pages.
Hierdie nasienriglyne bestaan uit 19 bladsye.**

NOTE:

- If a candidate answers a question TWICE, mark only the FIRST attempt.
- Consistent accuracy applies in all aspects of the marking guidelines where indicated.
- # Shows questions where a Tolerance Range will be applied:
Q 2.2 ; Q 5.2.1 ; Q 6.1 & Q 9.2

LET WEL:

- Indien 'n kandidaat 'n vraag TWEE keer beantwoord, sien slegs die EERSTE poging na.
- Volgehoue akkuraatheid is deurgaans op alle aspekte van die nasienriglyne soos aangedui.
- # Toon vrae waar Toleransie wydte (Verdraagsaamheids omvang) toegepas word:
V 2.2 ; V 5.2. ; V 6.1 & V 9.2

QUESTION/VRAAG 1

1.1.1	$x^2 - x - 12 = 0$ $(x - 4)(x + 3) = 0$ OR/OF $x = \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(-12)}}{2(1)}$ $x = 4$ or/of $x = -3$	✓ factors/formula faktore/formule ✓ both values of x beide waardes van x AO: Full Marks/Volpunte	A CA (2)
1.1.2	$x^2 - x - 12 \leq 0$ $(x - 4)(x + 3) \leq 0$ $-3 \leq x \leq 4$ OR/OF $x \in [-3 ; 4]$	✓ correct notation korrekte notasie ADDENDUM	CA (1)
1.1.3	$x^2 - x - 12 = -5$ $x^2 - x - 7 = 0$ $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ $= \frac{-(-1) \pm \sqrt{(-1)^2 - 4(1)(-7)}}{2(1)}$ $x \approx -2,19$ or/of $x \approx 3,19$	✓ standard form /standaardvorm ✓ SF ✓ positive x -value Positiewe x -waarde ✓ negative value of x negatiewe waarde v x NPR AO: Full Marks/Volpunte	A CA CA CA (4)

1.4	111110 ₂																				
	<table><tr><td>2⁵</td><td>2⁴</td><td>2³</td><td>2²</td><td>2¹</td><td>2⁰</td></tr><tr><td>1</td><td>1</td><td>1</td><td>1</td><td>1</td><td>0</td></tr><tr><td>32</td><td>16</td><td>8</td><td>4</td><td>2</td><td>0</td></tr></table>	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰	1	1	1	1	1	0	32	16	8	4	2	0		
	2 ⁵	2 ⁴	2 ³	2 ²	2 ¹	2 ⁰															
	1	1	1	1	1	0															
	32	16	8	4	2	0															
32 + 16 + 8 + 4 + 2 = 62	✓ 62																				
$\therefore 2 \left(111110_2 + 38 \right)$ $= 2 \left(62 + 38 \right)$ $= 200$	✓ 200 AO: Full Marks/Volpunte	A CA (2)																			
			[20]																		

QUESTION/VRAAG 2

2.1.1	$b = 0$	✓ Value of/ waarde van b	A (1)
2.1.2	$b = \frac{2}{5}$ OR / OF 0,4	✓ Value of/ waarde van b	A (1)
2.2 #	$kx^2 = 35 - 2x$ $kx^2 + 2x - 35 = 0$ $\Delta = b^2 - 4ac$ $= (2)^2 - 4(k)(-35)$ $= 4 + 140k$ For real roots / Vir reële wortels: $\Delta \geq 0$ $4 + 140k \geq 0$ $k \geq -\frac{1}{35}$ OR/OF $-0,03$	✓ standard form /standaardvorm ✓ SF ✓ S ✓ $\Delta \geq 0$ ✓ Values of /waardes van k ADDENDUM	A CA CA A CA (5)
			[7]

QUESTION/VRAAG 3

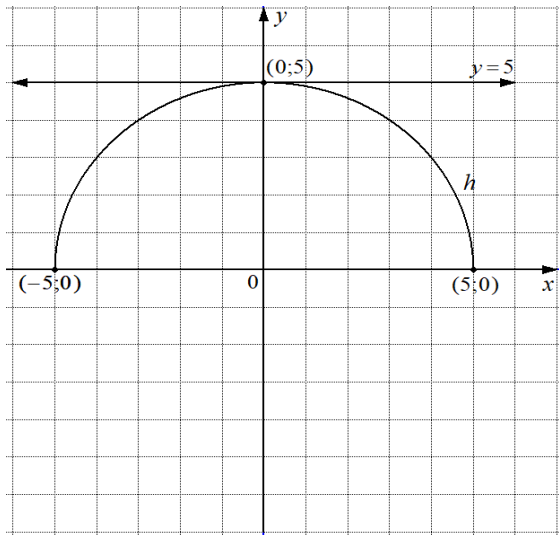
3.1.1	$\sqrt[3]{8x^{27}} = 2x^9$	✓ $2x^9$	A (1)
3.1.2	$9^{n+1} \times 4^n \times 6^{1-2n}$ $= (3^2)^{n+1} \times (2^2)^n \times (2 \times 3)^{1-2n}$ $= 3^{2n+2} \times 2^{2n} \times 2^{1-2n} \times 3^{1-2n}$ $= 3^{2n+2+1-2n} \times 2^{2n+1-2n}$ $= 3^3 \times 2^1$ $= 54$	✓ prime bases /priem grondtalle ✓ expansion/uitbreiding ✓ S	A CA CA (3)
3.1.3	$\sqrt{k}(2 - \sqrt{k}) - \sqrt{4k}$ $= 2\sqrt{k} - k - 2\sqrt{k}$ $= -k$ OR/OF $\sqrt{k}(2 - \sqrt{k}) - \sqrt{4k}$ $= k^{\frac{1}{2}} \left(2 - k^{\frac{1}{2}} \right) - 2k^{\frac{1}{2}}$ $= 2k^{\frac{1}{2}} - k - 2k^{\frac{1}{2}}$ $= -k$	✓ $2\sqrt{k} - k$ ✓ $-2\sqrt{k}$ ✓ S OR/OF ✓ $2k^{\frac{1}{2}} - k$ ✓ $-2k^{\frac{1}{2}}$ ✓ S	A A CA A A CA (3)
3.2.1	$\log 72 - \log 2$ OR/OF $\log 36 + \log 2 - \log 2 = \log 36$ $= \log \frac{72}{2}$ OR/OF $\log 36$ OR/OF $2 \log 6$	✓ log prop./eienskap	A (1)
3.2.2	$\frac{\log 72 - \log 2}{\log 6} = \frac{\log 36}{\log 6}$ $= \frac{\log 6^2}{\log 6}$ OR/OF $= \log_6 36$ $= \frac{2 \log 6}{\log 6}$ $= 2 \log_6 6$ $= 2$	AO: Full Marks if Q 3.2.1 is correct otherwise only 1mark <i>/Volpunte indien V3.2.1 korrek is anders Slegs 1 punt</i> ✓ log prop./eienskap ✓ S ADDENDUM	CA CA (2)

3.3	$5^{x+2} - 5^x = 600$ $5^x \cdot 5^2 - 5^x = 600$ $5^x (25 - 1) = 600$ $5^x = 25$ $5^x = 5^2$ OR/OF $x = \log_5 25$ OR/OF $\log 5^x = \log 5^2$ $x \log 5 = 2 \log 5$ $\therefore x = 2$	✓ separating exp./skei eksp. ✓ factors / faktore ✓ S ✓ exp or log. prop./eksp. of log eienskap ✓ value of /waarde van x AO: Full Marks/Volpunte	A CA CA CA CA CA (5)
3.4.1	$-i$ OR/OF $0 - i$	✓ conjugate /gekojugeerde	A (1)
3.4.2	$\frac{2+3i}{i}$ $= \frac{2+3i}{i} \times \frac{-i}{-i}$ $= \frac{-2i-3i^2}{-i^2}$ OR / OF $-\frac{2i}{-i^2} + 3$ $= \frac{-2i-3(-1)}{-(-1)}$ OR / OF $-\frac{2i}{-(-1)} + 3$ $= -2i + 3$	✓ M ✓ S ✓ -1 ✓ S ADDENDUM	A CA A CA (4)
3.5	$a + bi = -i - 14$ $\therefore a = -14$ and / en $b = -1$	✓ a-value/waarde ✓ b-value/waarde	A A (2)
			[22]

QUESTION/VRAAG 4

4.1.1	(3;18)	✓ 3 ✓ 18	A A (2)
4.1.2	$-2(x-3)^2 + 18 = 0$ $-2(x^2 - 6x + 9) + 18 = 0$ $-2(x^2 - 6x + 9) + 18 = 0$ $-2x^2 + 12x = 0$ $-2x(x-6) = 0$ OR/OF $x = \frac{-(12) \pm \sqrt{(12)^2 - 4(-2)(0)}}{2(-2)}$ $x = 0$ or / of $x = 6$ OR/OF $-2(x-3)^2 + 18 = 0$ $2(x-3)^2 = 18$ OR/OF $(x-3)^2 = 9$ $x-3 = \pm 3$ $x = 0$ or / of $x = 6$	✓ = 0 ✓ Standard form <i>standaard vorm</i> ✓ factors/formula <i>faktore/formule</i> ✓ both values of x <i>albei waardes van x</i> OR/OF ✓ = 0 ✓ S ✓ S ✓ both x -values of / <i>albei</i> <i>x-waardes</i>	A CA CA CA A CA CA CA (4)
4.1.3 & 4.1.5		4.1.3 f : ✓ Turning point <i>Draaipunt</i> ✓ both x - int <i>beide x-afsn.</i> ✓ shape/vorm ADDENDUM	CA CA CA (3)
		4.1.5 h : ✓ x - int./afsn. & (5;10) ✓ increasing line/ <i>toenemende lyn</i> ADDENDUM	CA CA (2)

4.1.4 (a)	$f(x) = -2(x-3)^2 + 18$ subst./verv. (5; t) $t = -2(5-3)^2 + 18$ $= 10$ OR/OF $f(x) = -2x^2 + 12x$ $t = -2(5)^2 + 12(5)$ $= 10$	✓ subst./verv ✓ value of /waarde van t OR/OF ✓ subst./verv ✓ value of /waarde van t AO: Full Marks/Volpunte	A CA CA CA (2)
4.1.4 (b)	$h(x) = 2x + c$ subst./verv (5; 10) $10 = 2(5) + c$ $c = 0$	✓ subst./verv ✓ c-value of /c-waarde	CA CA (2)
4.1.5	REFER TO DIAGRAM IN Q 4.1.3		
4.2.1 (a)	$x \in \square$; $x \neq 0$ OR/OF $x \in (-\infty; 0) \cup (0; \infty)$	✓ $x \neq 0$ OR/OF $x \in (-\infty; 0) \cup (0; \infty)$	A (1)
4.2.1 (b)	$y > -4$ OR/OF $y \in (-4; \infty)$	✓ Range/waarde versameling	A (1)
4.2.1 (c)	$q = -4$	✓ $q = -4$	A (1)
4.2.1 (d)	$D(0; -3)$	✓ $x = 0$ ✓ $y = -3$	A CA (2)
4.2.2	$0 = -\frac{8}{x} - 4$ $4 = -\frac{8}{x}$ $4x = -8$ $x = -2$ $C(-2; 0)$	✓ $y = 0$ ✓ S ✓ x-value /x-waarde AO: Full Marks/Volpunte	A CA CA (3)
4.2.3	$g(x) = a^x + q$ $g(x) = a^x - 4$ $0 = a^{-2} - 4$ subst./ver v. (-2; 0) $4 = a^{-2}$ $a = \frac{1}{2}$	✓ subst./verv ✓ a-value /a-waarde	CA CA (2)

4.3		<i>h:</i> ✓ shape/vorm ✓ x-intercepts / afsnitte ✓ line/lyn $y = 5$	A A CA (3)
			[28]

QUESTION/VRAAG 5

5.1.1	250	✓ 250	A (1)
5.1.2	$A = P(1 + i)^n$ $= 250(1 + 50\%)^{12}$ $\approx 32\,437$	✓F ✓SF ✓S ADDENDUM	A CA CA (3)
5.1.3	$A = P(1 + i)^n$ $100\,000 = 250(1 + 50\%)^n$ $\frac{100\,000}{250} = (1,5)^n$ $n = \frac{\log\left(\frac{100\,000}{250}\right)}{\log(1,5)}$ OR/OF $n = \log_{1,5} \frac{100\,000}{250}$ $\therefore n \approx 15 \text{ months/ maande}$	✓F ✓SF ✓ log form/-vorm ✓ value of/waarde van n ADDENDUM	A CA CA CA (4)
5.2.1 #	Value of investment end of 3 years/ <i>Waarde van belegging einde van 3 jaar:</i> $A = P(1 + i)^n$ $= R15\,000 \left(1 + \frac{8,5\%}{4}\right)^{3 \times 4}$ $\approx R19\,305,28$ Value of investment end of 5 years/ <i>Waarde van belegging einde van 5 jaar:</i> $A = P(1 + i)^n$ $= R19\,305,28 \left(1 + \frac{6\%}{2}\right)^{2 \times 2}$ $\approx R21\,728,26$ $\therefore R21\,728,26 < R23\,000$ He will NOT have enough money/ <i>Hy sal NIE</i> <i>genoeg geld hê NIE.</i>	 ✓ values of i and n / <i>waarde van i en n</i> ✓S ✓ values of i and n / <i>waarde van i en n</i> ✓ R21 728,26 ✓ conclusion/gevolgtrekking	 A CA A CA CA

	OR/OF $A = P(1+i)^n$ $= R15000 \left(1 + \frac{8,5\%}{4}\right)^{3 \times 4} \times \left(1 + \frac{6\%}{2}\right)^{2 \times 2}$ $\approx R 21\,728,26$ $\therefore R 21\,728,26 < R 23\,000$ He will NOT have enough money/Hy sal NIE genoeg geld hê NIE	OR/OF ✓ M ✓ values of <i>i</i> and <i>n</i>/ waarde van <i>i</i> en <i>n</i> ✓ values of <i>i</i> and <i>n</i>/ waarde van <i>i</i> en <i>n</i> ✓ R 21 728,26 ✓ conclusion/gevolgtrekking ADDENDUM	A A A CA CA (5)
5.2.2	$R 21\,728,26 - R15\,000$ $= R 6\,728,26$	✓ M subtracting / <i>aftrek</i> ✓ interest earned <i>rente verdien</i>	A CA (2)
			[15]

6.3	$g(x) = -\frac{9}{x}$ $g(-3) = -\frac{9}{(-3)} = 3$ $g(-1) = -\frac{9}{(-1)} = 9$ $\text{Av./Gemid. Gradient} = \frac{g(x_2) - g(x_1)}{x_2 - x_1}$ $= \frac{3 - 9}{-3 - (-1)} \quad \text{OR/OF} \quad = \frac{9 - 3}{-1 - (-3)}$ $= 3$	✓ both $g(x)$-values beide $g(x)$-waardes ✓ F ✓ 3	A A CA (3)
6.4.1	$f(x) = mx^3 + mx - 4$ $\therefore f'(x) = 3mx^2 + m + 0$	✓ $3mx^2$ ✓ $m + 0$ OR/OF $m - 0$	A A (2)
6.4.2	$f'(2) = 3m(2)^2 + m = 13m$	✓ $13m$	CA (1)
6.4.3	$f'(2) = 39$ $13m = 39$ $m = 3$	✓ Equating derivative to 39/ Stel afgeleide gelyk aan 39 ✓ Value of / waarde van m	CA CA (2)
			[20]

QUESTION/VRAAG 7

7.1	$F(0; -2)$	✓ 0 ✓ -2	A A (2)
7.2	$2^3 + p(2)^2 + 9(2) - 2 = 0$ $8 + 4p + 18 - 2 = 0$ OR/OF $4p = -24$ $p = -6$	✓ Subst./verv. (2; 0) ✓ S	A A (2)
7.3	x-intercepts /afsnitte; $y = 0$ $(x - 2)(x^2 - 4x + 1) = 0$ $\therefore x = 2$ or/of $x = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(1)}}{2(1)}$ $\therefore x_B = 2$ and / en $x_A = 2 - \sqrt{3}$ and / en $x_C = 2 + \sqrt{3}$ $\therefore BC = \sqrt{3}$	✓ = 0 ✓ quadratic factor kwadratiese faktor ✓ SF ✓ values of/waardes van x ✓ length of / lengte van BC AO: Full Marks/Volpunte ADDENDUM	A A CA CA CA (5)
7.4	$h'(x) = 3x^2 - 12x + 9 = 0$ $3(x - 1)(x - 3) = 0$ OR/OF $x = \frac{-(-12) \pm \sqrt{(-12)^2 - 4(3)(9)}}{2(3)}$ $\therefore x = 1$ or/of $x = 3$ $h(1) = (1)^3 - 6(1)^2 + 9(1) - 2 = 2$ $h(3) = (3)^3 - 6(3)^2 + 9(3) - 2 = -2$ $\therefore D(1; 2)$ and / en $E(3; -2)$	✓ derivative/afgeleide ✓ derivative = 0 afgeleide = 0 ✓ factors/formula faktore/formule ✓ both x-values beide x-waardes ✓ both y-values beide y-waardes AO: Full Marks/Volpunte ADDENDUM	A A CA CA CA (5)

7.5	$2 < x < 3$ or / of $x > 2 + \sqrt{3}$ OR/OF $x \in (2; 3)$ or / of $x \in (2 + \sqrt{3}; \infty)$ OR/OF $x > 2$ and /en $x < 3$ or / of $x > 2 + \sqrt{3}$	✓ critical values kritiese waardes ✓ correct notation korrekte notasie ✓ $x > 2 + \sqrt{3}$ OR/OF ✓ critical values kritiese waardes ✓ correct notation korrekte notasie ✓ $x \in (2 + \sqrt{3}; \infty)$ OR/OF ✓ critical values kritiese waardes ✓ correct notation korrekte notasie ✓ $x > 2 + \sqrt{3}$	CA A CA CA CA CA CA A CA (3)
			[17]

QUESTION/VRAAG 8

8.1	$h(0) = -(0)^2 + 6(0) + 1,62$ $= 1,62 \text{ m}$	✓ 1,62 m	A NPU (1)
8.2	$h'(t) = -2t + 6$	✓ $-2t + 6$	A (1)
8.3	$h'(t) = 0 \quad \text{OR/OF} \quad t = -\frac{b}{2a}$ $\therefore -2t + 6 = 0 \quad t = -\frac{(6)}{2(-1)}$ $t = 3 \text{ s}$ $h(3) = -(3)^2 + 6(3) + 1,62$ $= 10,62 \text{ m}$ OR/OF $h = \frac{4ac - b^2}{4a}$ $h = \frac{4(-1)(1,62) - (6)^2}{4(-1)}$ $= 10,62 \text{ m}$	✓ derivative/ <i>afgeleide</i> = 0 OR/OF using/ <i>gebruik</i> $-\frac{b}{2a}$ ✓ <i>t</i> -value / <i>waarde</i> ✓ Subst. <i>t</i> -value/ <i>waarde</i> ✓ Maximum height/ <i>maks hoogte</i> OR/OF ✓ F ✓✓ subst/verv. <i>a, b</i> and/en <i>c</i> <i>value/ waarde</i> ✓ Maximum height/ <i>maks hoogte</i>	A CA CA CA NPU A A A CA NPU (4)
8.4	$-2t + 6 = 3$ $t = 1,5 \text{ s}$ $\therefore h(1,5) = -(1,5)^2 + 6(1,5) + 1,62$ $\approx 8,37 \text{ m}$ OR/OF $-2t + 6 = -3$ $t = 4,5 \text{ s}$ $\therefore h(4,5) = -(4,5)^2 + 6(4,5) + 1,62$ $\approx 8,37 \text{ m}$	✓ derivative / <i>afgeleide</i> = 3 ✓ <i>t</i> -value / <i>waarde</i> ✓ height / <i>hoogte</i> OR/OF ✓ derivative / <i>afgeleide</i> = -3 ✓ <i>t</i> -value / <i>waarde</i> ✓ height / <i>hoogte</i>	CA CA CA NPU CA CA CA NPU (3)
			[9]

QUESTION/VRAAG 9

9.1.1	$\int x^4 dx$ $= \frac{x^5}{5} + C$	$\checkmark \frac{x^5}{5}$ $\checkmark C$	A A (2)
9.1.2	$\int \left(2\pi + \frac{4}{x} \right) dx$ $= 2\pi x + 4 \ln x + C$	$\checkmark 2\pi x$ $\checkmark 4 \ln x + C$ No Penalty if C omitted/ Geen penalisering indien C weggelaat	A A (2)
9.1.3	$\int (2x^{-3})^3 dx$ $\int 8x^{-9} dx$ $= -x^{-8} + C \quad \text{OR/OF} \quad = -\frac{1}{x^8} + C$	$\checkmark 8x^{-9}$ $\checkmark -x^{-8} + C$ No Penalty if C omitted/ Geen penalisering indien C weggelaat	A CA (2)
9.2 #	$A = \int_{-1}^3 g(x) dx$ $= \int_{-1}^3 (2^x + 2) dx$ $= \left[\frac{2^x}{\ln 2} + 2x \right]_{-1}^3$ $= \left[\frac{(2)^3}{\ln 2} + 2(3) \right] - \left[\frac{(2)^{-1}}{\ln 2} + 2(-1) \right]$ $\approx 18,82 \text{ units}^2 / \text{eenhede}^2$ OR/OF $= 8 + \frac{15}{2 \ln 2} \text{ units}^2 / \text{eenhede}^2$	$\checkmark$ Area notation using integrals/ <i>Oppervlak-notasie</i> <i>met gebruik van integrale</i> $\checkmark \frac{2^x}{\ln 2}$ $\checkmark + 2x$ $\checkmark \checkmark \text{ SF}$ $\checkmark \text{ area /oppervl}$ AO: 1 mark/ 1 punt NPU No substitution or simplification marks to be allocated if there is no integration/ Geen vervangings- of vereenvoudigingspunte moet toegeken word as daar geen integrasie is nie. ADDENDUM	M A A CA CA (6)
			[12]

TOTAL/TOTAAL: 150